

Helix Thready — API & SDKs (Area Index)

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Rev	Date	Author	Change
1	2026-07-21	swarm (API & SDKs)	Initial draft: OpenAPI 3.1 surface, event contract, authn/authz, error model, versioning, SDK strategy
2	2026-07-21	swarm (API & SDKs)	Second-pass review: added contract-tests.md (TDD, 15 test types); OpenAPI <code>x-thready-maturity</code> + <code>hmacAuth</code> + full oauth2 scopes + 422 + idempotency; p95 SLO; API-owned DDL (idempotency ledger, event sinks)
4	2026-07-22	swarm (API & SDKs)	Completeness-critic pass. Recorded the <code>#8</code> VectorDB (pgvector-only; Qdrant/Pinecone/Milvus swap-by-config, P1) gap in the coverage table and at <code>/search</code> ; fixed a stale <code>[OPEN: evt-2]</code> /“deferred” note left inside <code>openapi.yaml</code> ’s <code>eventDelivery</code> webhook (the <code>/v1/event-sinks</code> CRUD already ships in the same file); added the missing retry-poller partial index to the <code>event_sink_delivery</code> DDL.
3	2026-07-22	swarm (API & SDKs)	Pass-3 depth (enterprise-complete). Added <code>asynccapi.yaml</code> (AsyncAPI 3.0, one message per event type) and <code>sdk-examples.md</code> (7 recipes × 6 languages). Expanded <code>openapi.yaml</code> (Rev 3): <code>GET /version</code> (closes ver-2), <code>/v1/event-sinks</code> CRUD (closes evt-2), request/response examples, fail-loud/lost-race error examples — now 46 paths / 59 operations / 40 schemas. Re-verified auth+eventbus at source: corrected the RS256 story (<code>jwt.Config.Secret []byte</code> is HMAC-only; no <code>SetKey</code>), <code>APIKeyHeader(validator,</code>

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			<code>headerName)</code> , <code>oauth</code> <code>AutoRefresher</code> defaults; documented that the NATS module gives ephemeral <code>DeliverNew</code> + <code>LimitsPolicy</code> (no native replay/sticky) so those are BUILD-NEW. Closed authz-2 (kid scheme).

This area specifies the **externally-facing contract** of Helix Thready: the REST `/v1` surface (OpenAPI 3.1), the real-time event subscription contract (WebSocket/SSE + sticky events), authentication/authorization, the canonical error model, URL-path versioning, and the multi-language SDK generation strategy. It is the implementation-ready contract that clients, SDKs and integration tests build against.

All content obeys **CONVENTIONS.md** and never contradicts the authoritative sources (the final answered request, the subsystem gap register). Facts tagged `[VERIFIED]` were read at module source; proposals are tagged `[DEFAULT – adjustable]`; every relevant gap-register item is addressed inline with a `[GAP: id]` tag.

Upstream / Downstream dependencies

Upstream (this area consumes):

- **architecture/** — component boundaries, event-driven workflow (§2, §3 of the final request), concurrency/idempotency model. The API surface is a projection of those components.
- **database/** — the relational schema (posts, threads, replies, hashtags, categories, assets, `asset_links`, `processing_state`, `skills`, `accounts`, `users`, `roles`, `memberships`, `events`, `subscriptions`, `metering`, `audit`) backs the response/request schemas here 1:1. The vector store (pgvector) backs `/v1/search`.
- **In-house modules** (decision matrix): `digital.vasic.auth` (authn/authz), `digital.vasic.eventbus` (event envelope + NATS JetStream), `digital.vasic.http3` (transport), `digital.vasic.middleware` / `ratelimiter` / `security` (edge chain), `vasic-digital/helix_proto` (the codegen pattern), `HelixDevelopment/helix_skills`, `vasic-digital/herald`, `Catalogizer`. Real interfaces were read at source.

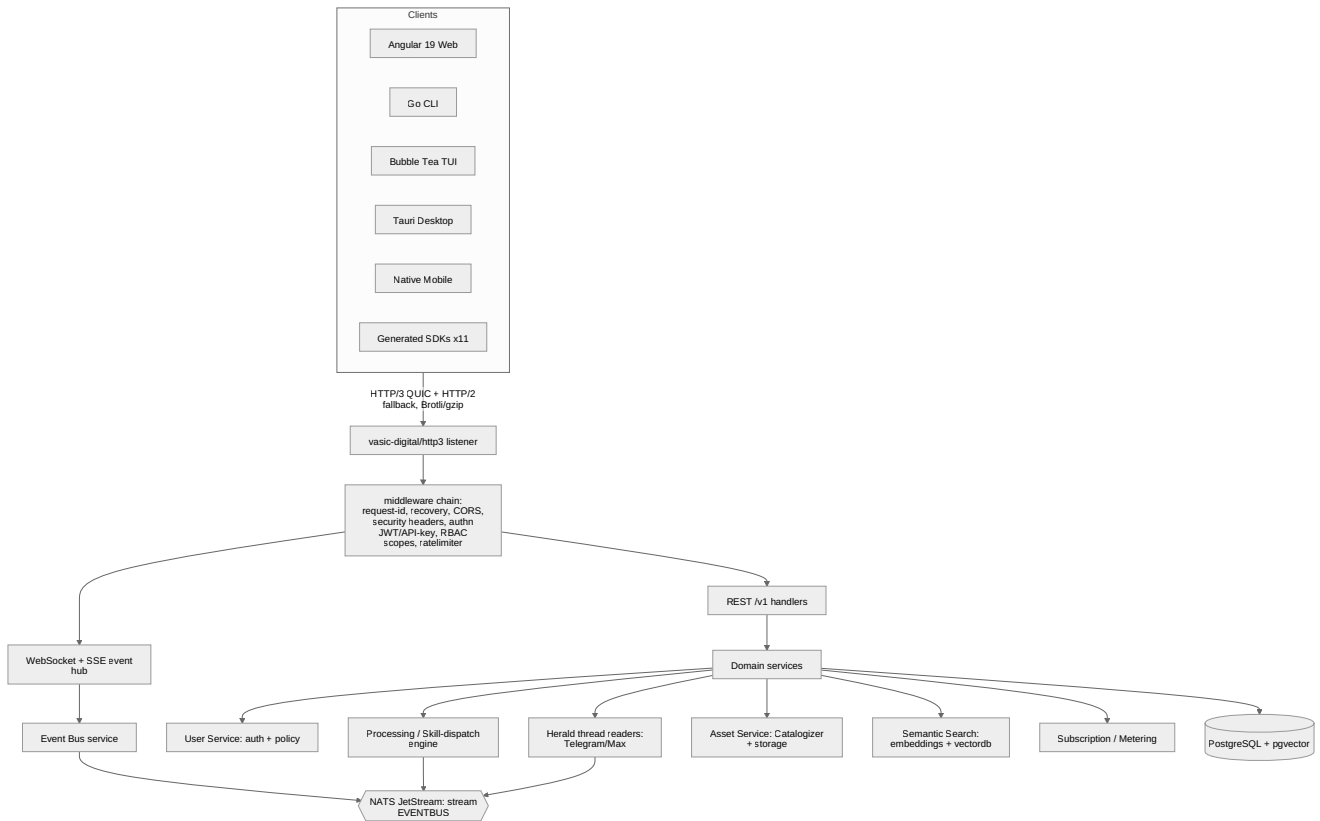
Downstream (this area feeds):

- **design/** and **user-guides/** — screens and CLI/TUI commands consume these endpoints and events.
- **testing/** — consumes the reproduce-first RED skeletons in `contract-tests.md`: the OpenAPI/Protobuf round-trip gate and the 15 mandated test types target this surface.
- The 11 **generated SDKs** are produced from `openapi.yaml` + the Protobuf contract.

Files in this area

File	Purpose
openapi.yaml	The OpenAPI 3.1 contract for the REST <code>/v1</code> surface (46 paths, 59 operations, 40 schemas, webhooks, per-op examples). Schema-first single source of truth.
asyncapi.yaml	The AsyncAPI 3.0 contract for the real-time plane: WS/SSE channels, outbound-webhook + inbound-callback channels, one message per event type with typed payload schemas. Sibling to <code>openapi.yaml</code> .
rest-endpoints.md	Endpoint-by-endpoint behaviour: resource groups, pagination, idempotency, filtering, rate limits, and backing-service maturity per group.
event-bus-contract.md	WebSocket/SSE subscription contract: event envelope, event catalog, subscription patterns, sticky events + invalidation, reconnect/replay, outbound webhooks, and the verified module-reality gap (§2a).
authn-authz.md	JWT access+refresh (RS256/EdDSA + JWKS), scoped API keys, OAuth2 linking, three-tier RBAC — grounded in and re-verified against <code>digital.vasic.auth</code> .
error-model.md	Canonical error envelope, the stable code table (1:1 with Protobuf/Connect codes), retry semantics, and the request-lifecycle failure map.
versioning.md	URL-path <code>/v1</code> versioning, additive-vs-breaking policy, <code>buf</code> breaking gate, deprecation/Sunset, proto-package parity, <code>GET /version</code> .
sdk-strategy.md	OpenAPI 3.1 + Protobuf codegen via the <code>helix_proto</code> pattern (<code>buf</code> → Go/Rust, <code>openapi-generator</code> → TS/Dart), the 11 target languages, and publishing.
sdk-examples.md	Per-language usage recipes (7 recipes × Go/TS/Python/Kotlin/Swift/Rust): auth, cursor iteration, idempotent async, search + embedder guard, event subscribe/reconnect, typed errors, transparent JWT refresh.
contract-tests.md	TDD reproduce-first RED skeletons for the <code>/v1</code> contract, mapped to the 15 mandated test types [<code>CONSTITUTION §11.4.27</code>] and the local (no-server-CI) pre-tag gate.
diagrams/	Mermaid <code>.mmd</code> sources for every diagram in these docs.

Surface at a glance



Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: [diagrams/api-surface.mmd](#).

Explanation (for readers/models that cannot see the diagram). Every client — the Angular web portal, the Go CLI, the Bubble Tea TUI, the Tauri desktop app, the native mobile clients, and the eleven generated SDKs — reaches the system over a single edge listener provided by `vasic-digital/http3`, which serves HTTP/3 (QUIC) with an HTTP/2 fallback and Brotli/gzip compression. Before any business logic runs, each request passes through a shared middleware chain (`digital.vasic.middleware` for request-id/recovery/CORS, `security/pkg/headers` for security headers, `digital.vasic.auth` for JWT/API-key authentication and RBAC scope checks, and `digital.vasic.ratelimiter` for quotas). Authenticated requests then split into two planes: the synchronous REST `/v1` handlers and the real-time WebSocket/SSE event hub.

The REST handlers delegate to domain services — the User Service (built on `digital.vasic.auth` + `security/pkg/policy`), the extended Herald thread readers for Telegram and Max, the processing / Skill-dispatch engine, the Asset Service (Catalogizer + `digital.vasic.storage`), the in-house semantic search (`embeddings` + `vectordb/pgvector`), and the subscription/metering service. These services persist to a single PostgreSQL instance that also hosts pgvector, keeping the datastore count low. The event hub is fed by the Event Bus service, a thin client-facing wrapper over `digital.vasic.eventbus` and its NATS JetStream stream `EVENTBUS`; the Herald readers and the processing engine publish events

(e.g. `post.received`, `processing.completed`) onto that stream, which the hub fans out to subscribed clients.

Gap-register items addressed in this area

Gap	Where addressed
#10 / 7.2 auth JWT default HMAC-SHA256 → asymmetric + JWKS + RBAC + MFA	authn-authz.md
#20 / New User Service (three-tier multi- tenant RBAC)	authn-authz.md, rest-endpoints.md
#5 / New Event Bus service (client-facing JetStream subscription) + sticky events	event-bus-contract.md (incl. verified module-reality \$2a), asyncapi.yaml
#11 SDK generation (OpenAPI 3.1 + Protobuf, helix_proto pattern)	sdk-strategy.md, sdk-examples.md
#18 TS client libs / scaffolds (no CI, no deep audit)	sdk-strategy.md, sdk-examples.md
#1 HelixLLM HashEmbedder trap (search must use real embedder)	rest-endpoints.md /search
#6 / 4.1 helix_skills has no execution engine (dispatch is BUILD-NEW)	rest-endpoints.md processing
#3 / 5.1 Herald Telegram reader trapped in QA harness; Max = stub	rest-endpoints.md channels
New Semantic-search service (Lumen-style, in- house)	rest-endpoints.md /search
#8 VectorDB pgvector- only (Qdrant/Pinecone/ Milvus swap-by-config, P1)	rest-endpoints.md /search \$2.8/\$3
#6.6 / 6.5 standardized 3rd-party callback contract; MeTube webhook	event-bus-contract.md, rest-endpoints.md, contract-tests.md
	error-model.md, authn-authz.md, contract-tests.md

Gap	Where addressed
#14 DDoS/rate limiting at the edge	
§12 anti-bluff sweep (non-GA ops proven, not claimed)	contract-tests.md (<code>x-thready-maturity</code> guard + fail-loud negative controls)
CONVENTIONS §6 (TDD reproduce-first, 15 test types)	contract-tests.md

Open items (tracked)

See each file’s “Open items” section. Summary of what **remains open** after the Pass-3 depth sweep:

- `[OPEN: api-1]` The Protobuf event/DTO contract (`proto/helix/thready/v1/*.proto`) is specified structurally here (and mirrored in `asynccapi.yaml`) but the `.proto` files themselves are produced with the code, in the `helix_thready_proto` repo — tracked in `sdk-strategy.md`.
- `[OPEN: api-2]` Concrete rate-limit tiers per plan are placeholders pending the billing/metering pack — tracked in `error-model.md`.
- `[OPEN: api-3]` Zig has no first-class OpenAPI/Protobuf generator; SDK path is hand-written over the C ABI / REST — tracked in `sdk-strategy.md`.

Closed / advanced in this pass: `[RESOLVED: ver-2]` `GET /version` added to `openapi.yaml` ; `[RESOLVED: evt-2]` `/v1/event-sinks` CRUD added; `[RESOLVED: authz-2]` `kid` scheme + rotation cadence fixed (authn-authz §11); `[RESOLVED: rest-1]` per-endpoint examples now in `openapi.yaml` + `sdk-examples.md`. New tracked build items surfaced by source re-verification: `[BUILD-NEW: auth-rs256]` (extend `jwt.Config` for asymmetric keys) and the Event Bus service’s durable-consumer + last-value work (`event-bus-contract` §2a).

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